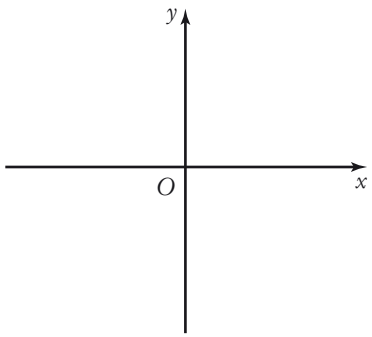
WORKSHEET

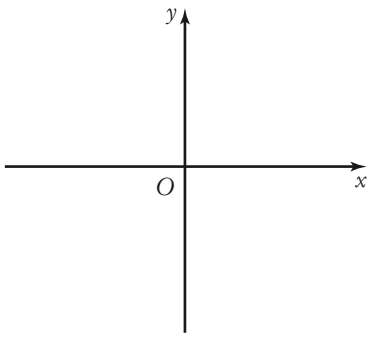
Chapter 4: Complex plane graphs

1 Sketch the circle defined by each equation below:

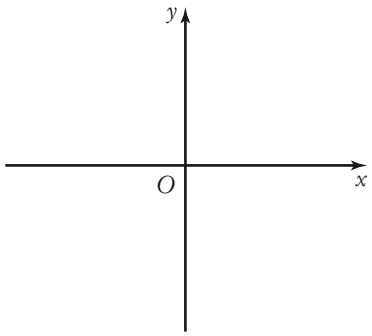
a $|z| = 3$



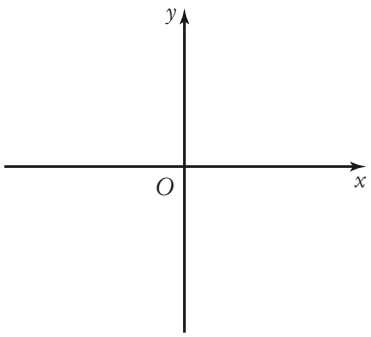
b $|z| = 7$



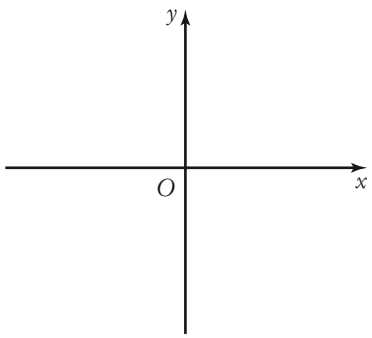
c $|z| = 3\sqrt{2}$



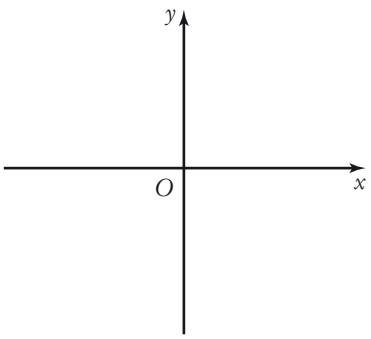
d $|z| = \frac{3}{4}$



e $|z - 2| = 4$

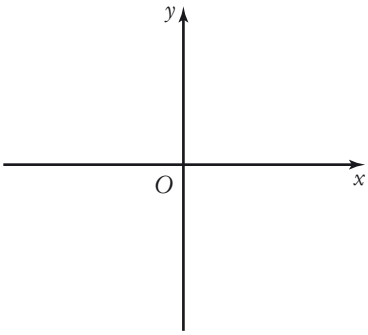


f $|z + 1| = 3$

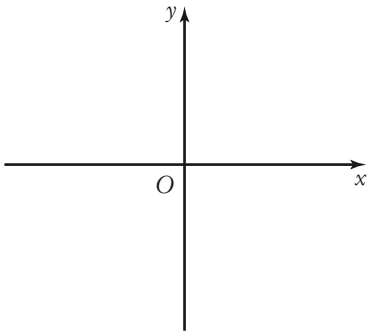


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g $|z - 2i| = 5$

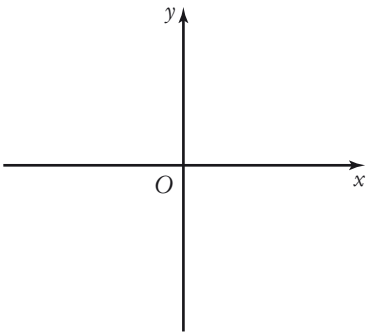


h $|z + 3i| = 4$

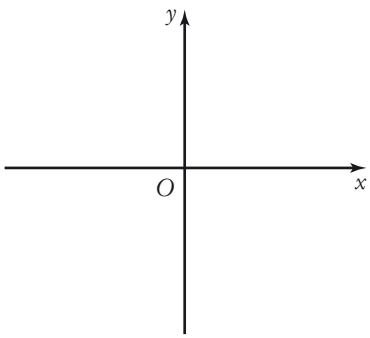


2 Sketch each of the following.

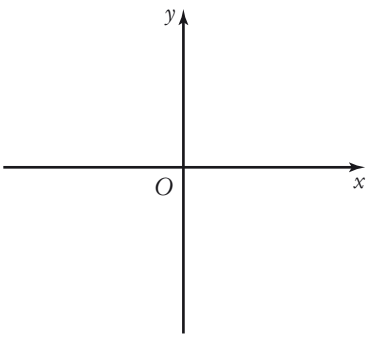
a $|z - (1 + i)| = 3$



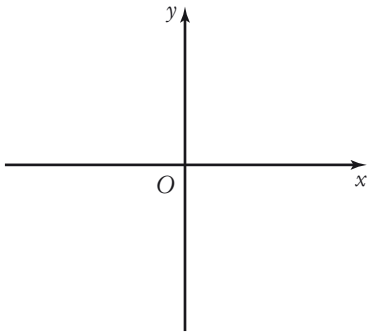
b $|z - (3 - 2i)| = 4$



c $|z - 1 - 3i)| = 2$



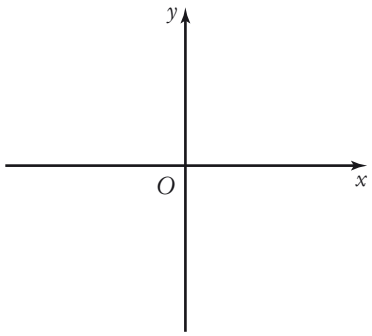
d $|z + 5 - 3i)| = 6$



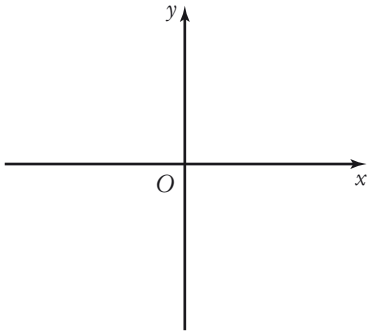
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3 Sketch each region defined by the inequalities below.

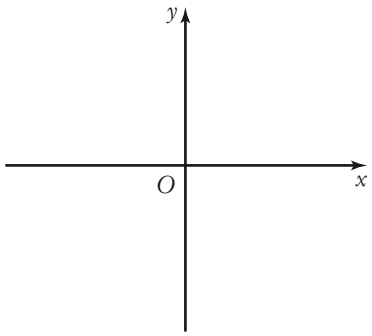
a $|z| \leq 5$



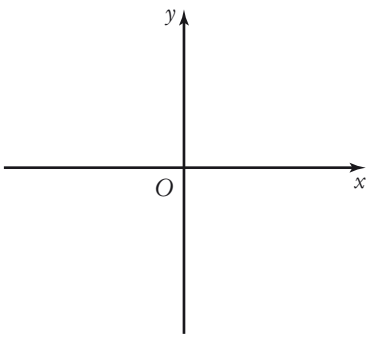
b $|z| > 2$



c $2 \leq |z| < 5$

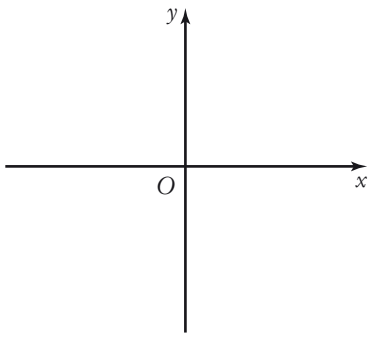


d $3 < |z + 2| \leq 6$

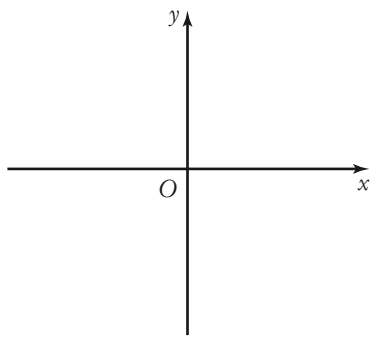


4 Sketch z for each of the following.

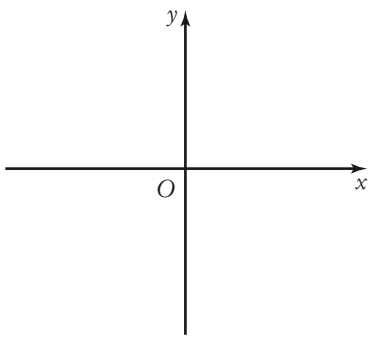
a $\text{Re}(z) = 4$



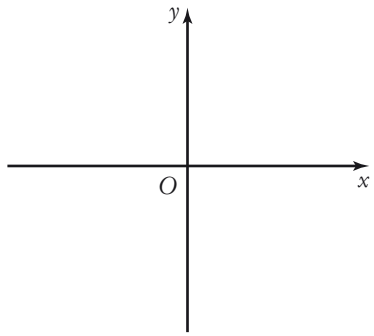
b $\text{Im}(z) = 7$



c $\text{Re}(z) = -5$



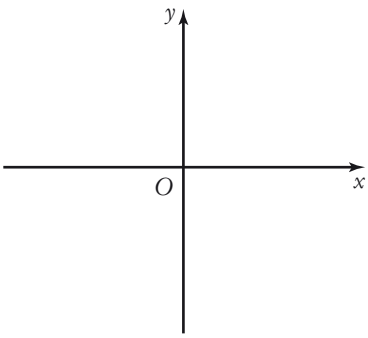
d $\text{Im}(z) = -6$



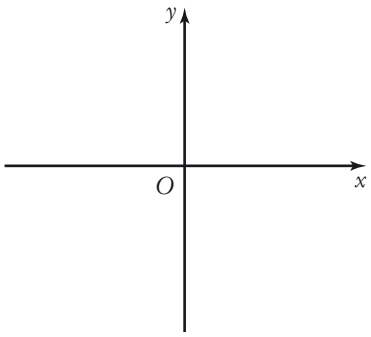
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5 Using the fact that $z = x + yi$ where x and y are real, find the Cartesian equation of each locus below and hence sketch.

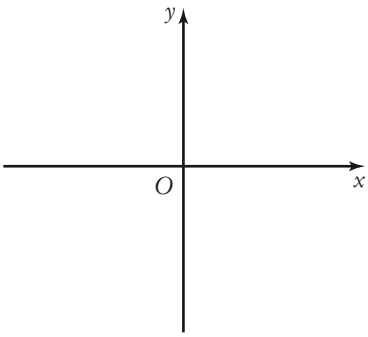
a $\text{Im}(z) = -\text{Re}(z)$



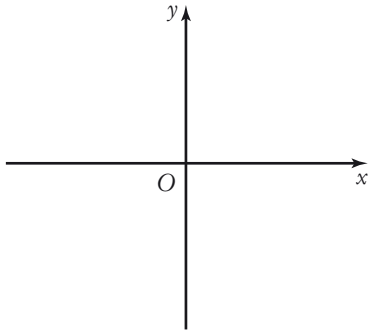
b $2\text{Im}(z) = \text{Re}(z)$



c $8\ \text{Re}(z) = 4\ \text{Im}(z) + 24$

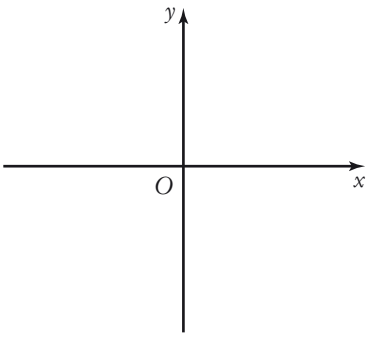


d $\text{Im}(z) + 3\ \text{Re}(z) = 9$

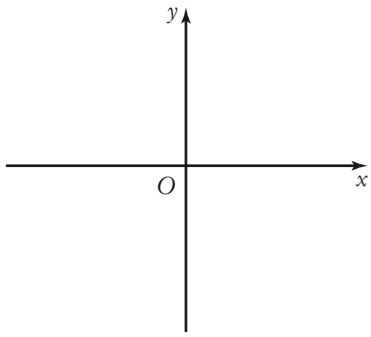


6 Sketch the parabola defined by each equation below.

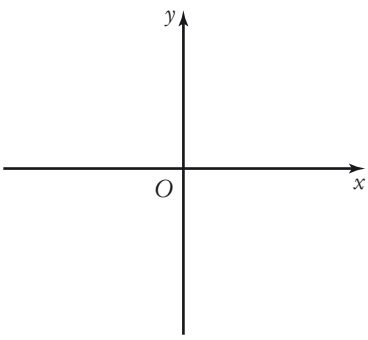
a $\text{Re}(z) = |z - 1 - i|$



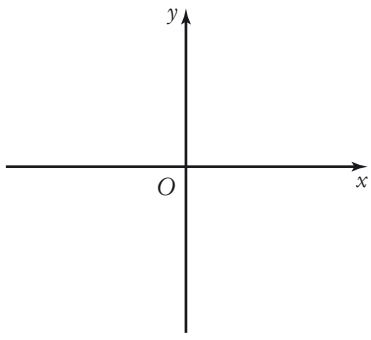
b $\text{Im}(z) = |z + 2 + 3i|$



c $|z - 2 + 2i| = \text{Re}(z)$



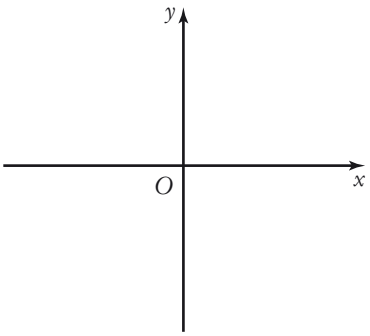
d $|z + 3 - 4i| = \text{Im}(z)$



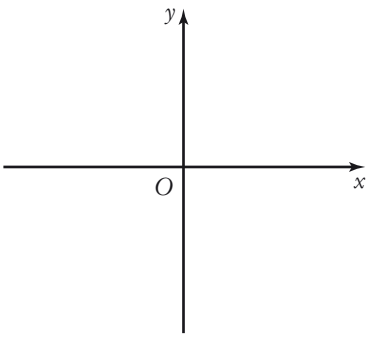
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7 Sketch z defined by each inequality below.

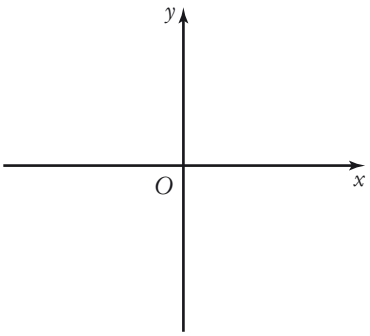
a $\text{Re}(z) < -5$



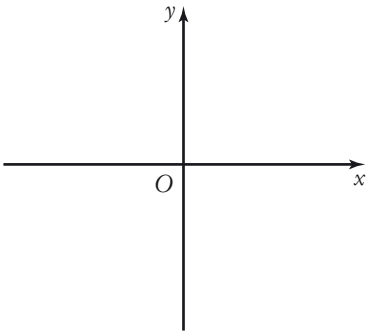
b $\text{Im}(z) > 2$



c $-3 \leq \text{Re}(z) < 2$

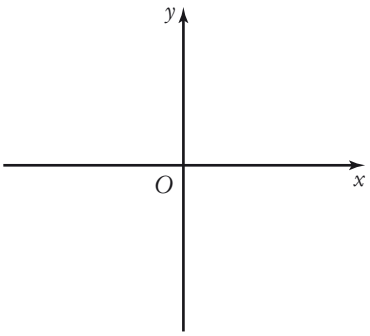


d $\text{Re}(z) < 5$ and $\text{Im}(z) \geq 2$

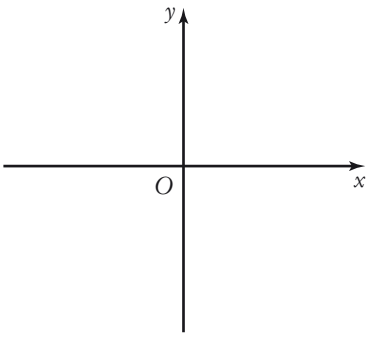


8 Describe each of the loci below and hence sketch.

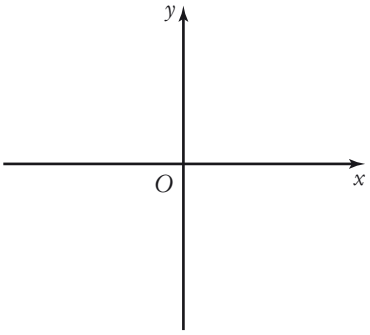
a $|z| = |z + 3i|$



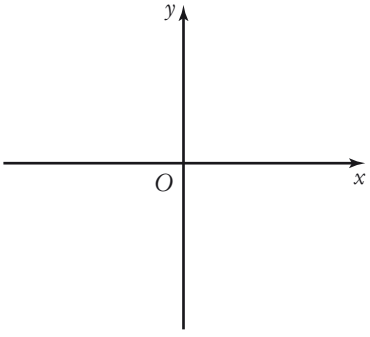
b $|z + 4| = |z - 2|$



b $|z + 2 - i| = |z + 2 + i|$

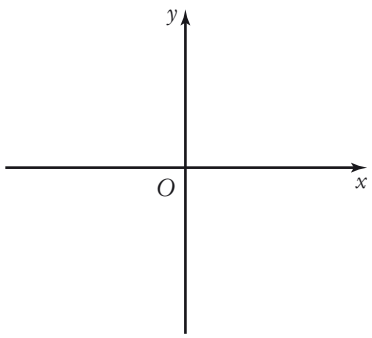


c $|z - 4 + i| = |z - 3 - 4i|$

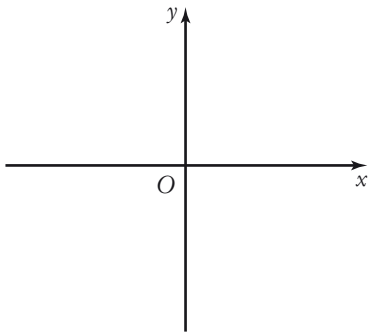


9 Sketch each of the following regions.

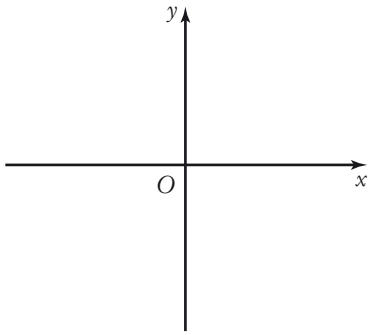
a $|z| > 3$ and $0 \leq \arg(z) \leq \frac{\pi}{4}$



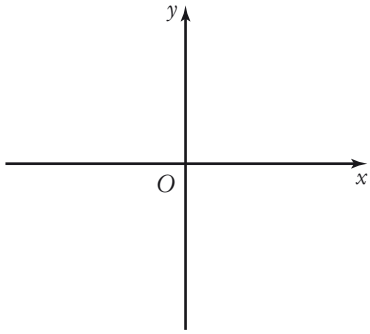
b $|z - 2| \leq 3$ and $0 < \arg(z) < \frac{\pi}{6}$



c $|z + 5i| \geq 2$ and $\frac{-5\pi}{6} \leq \arg(z) < \frac{-\pi}{6}$



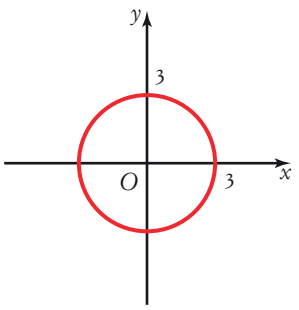
d $3 \geq |z| \leq 6$ and $\frac{-\pi}{4} < \arg(z) < \frac{3\pi}{4}$



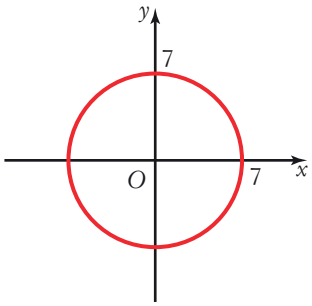
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Answers

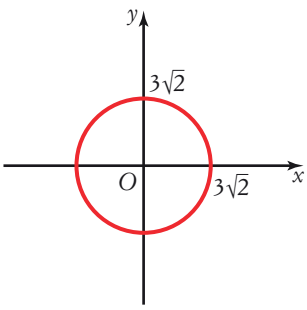
1 a



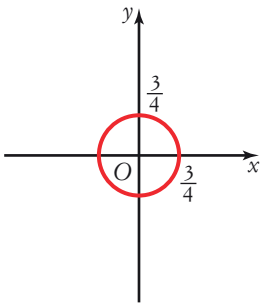
b



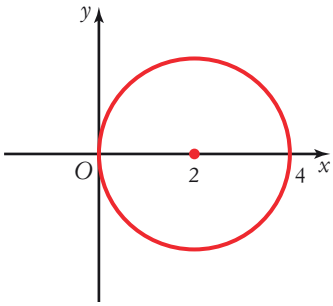
c



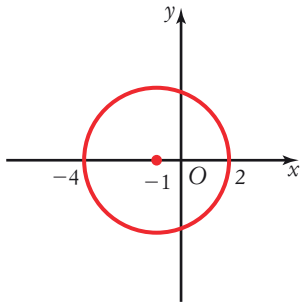
d



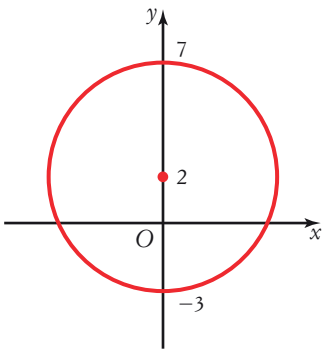
e



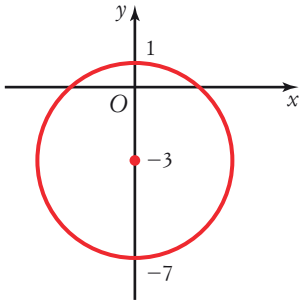
f



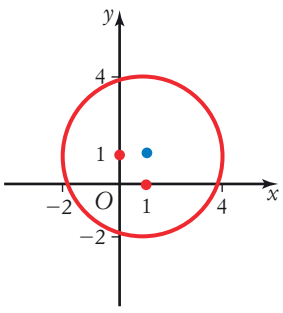
g



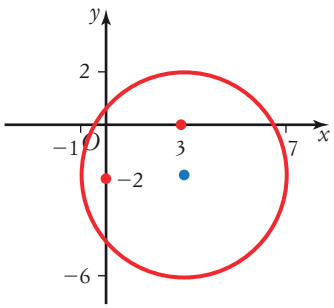
h



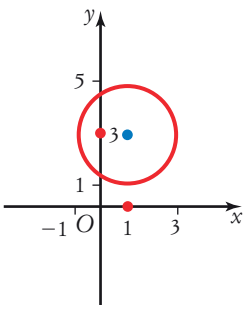
2 a



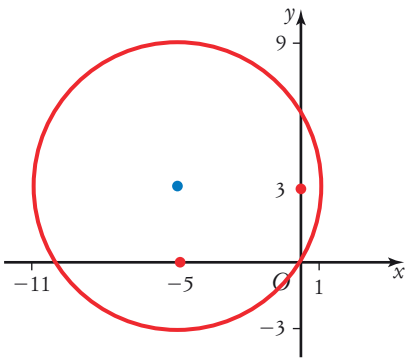
b



c

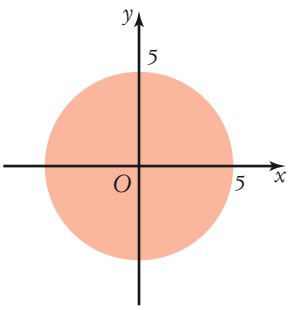


d

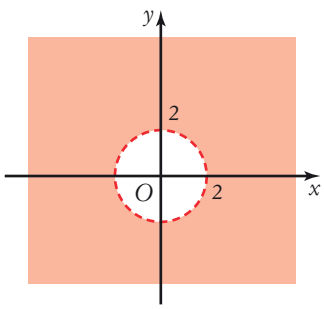


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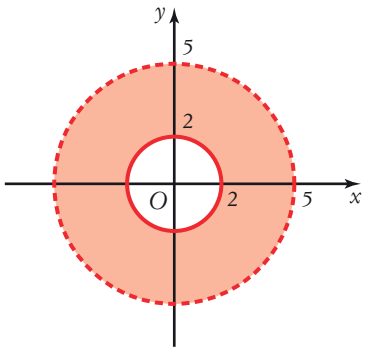
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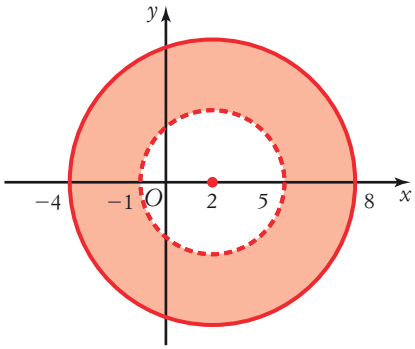
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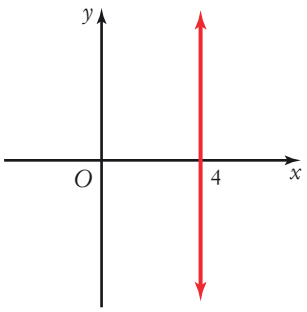
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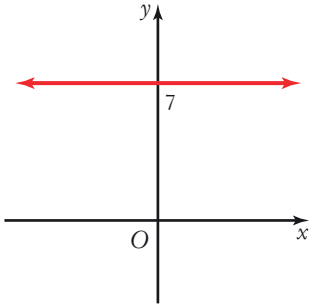
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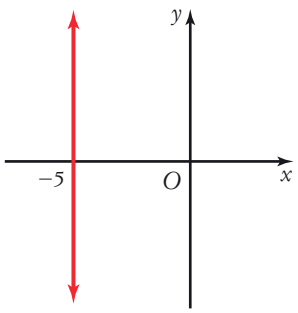
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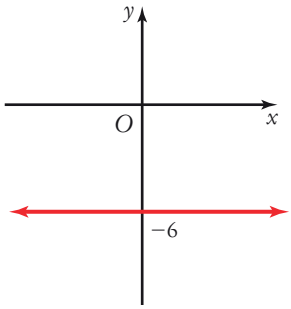
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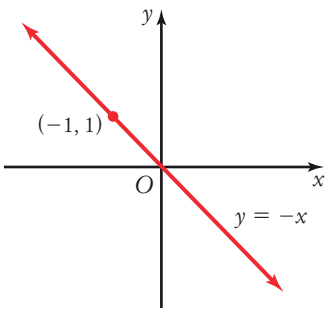


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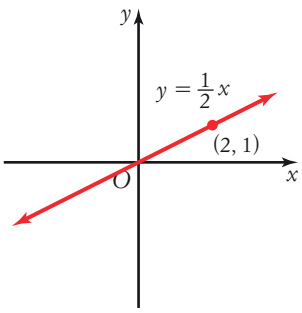


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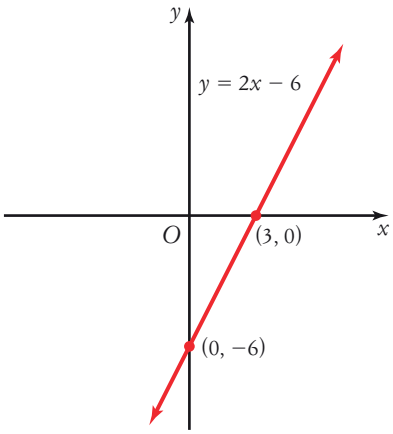
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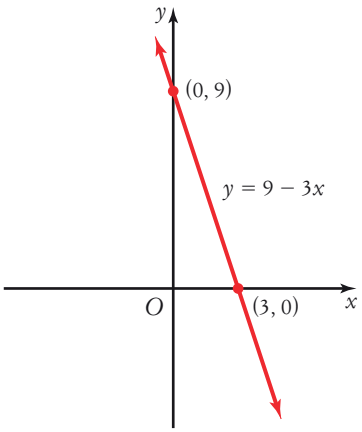
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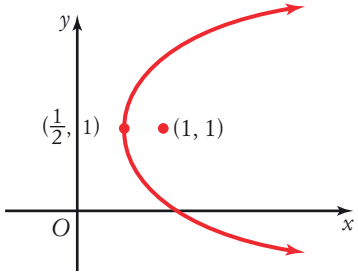
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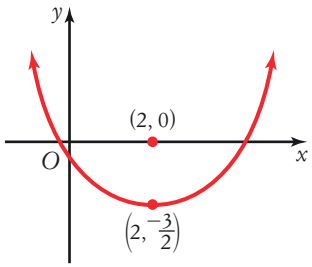
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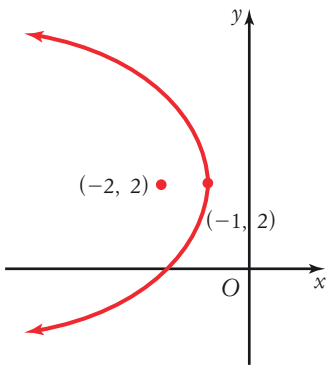
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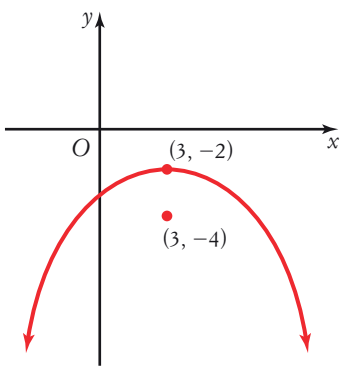
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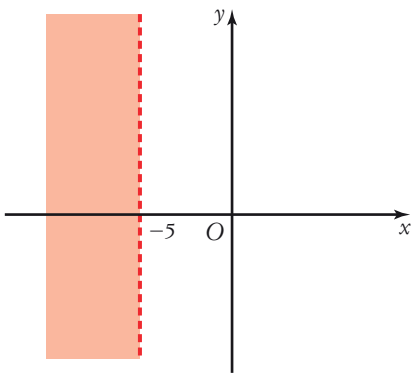


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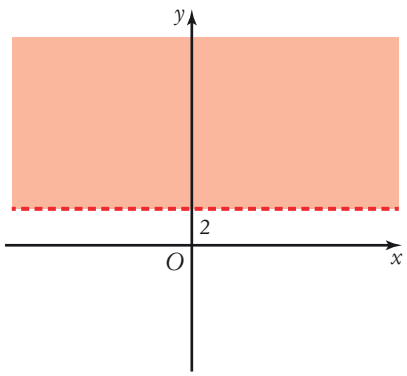


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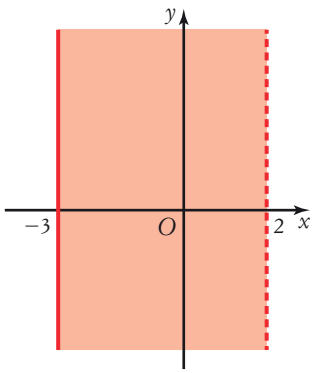
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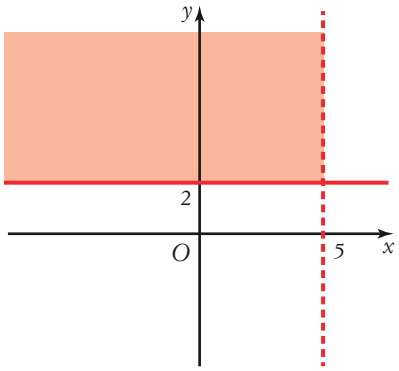
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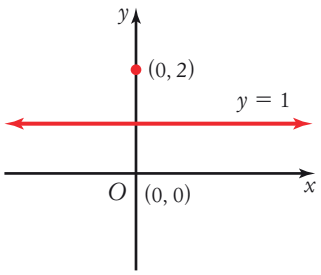
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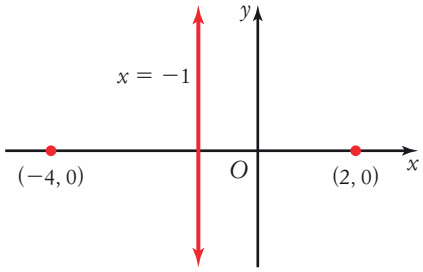
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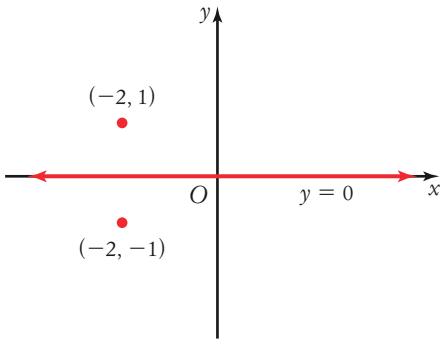
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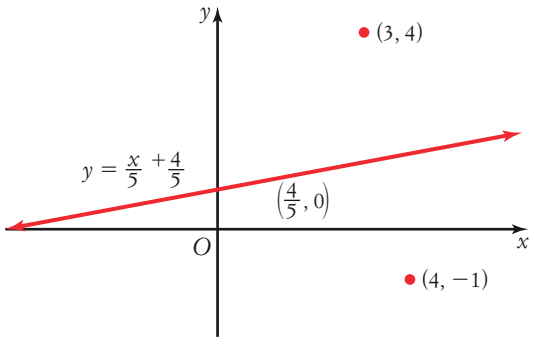
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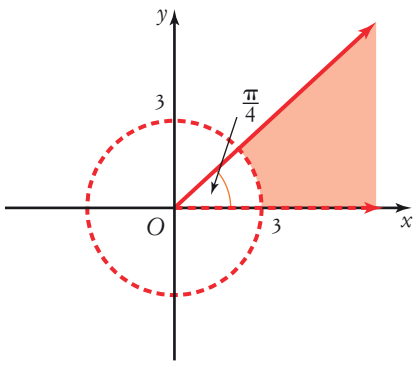


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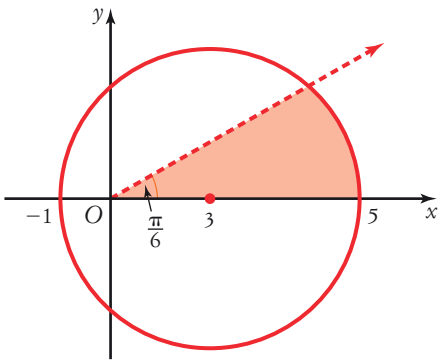


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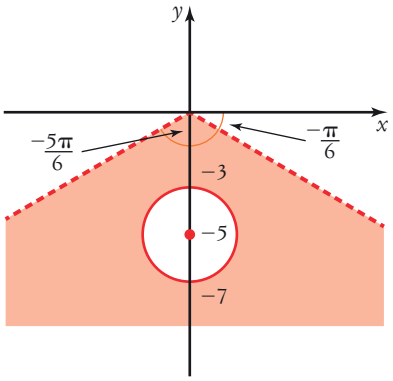
9 a



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